

WEATHER PERMITTING

DESCRIPTION: This event will test the team's knowledge of conducting investigations and using appropriate technology to build an understanding of weather.

COMPETITORS BRING:

- Pencil/writing utensil
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

NUMBER OF PARTICIPANTS: Maximum of 2

IMPOUND: NO

THE COMPETITION:

1. The event may be run as a sit-down test, station or a combination of the two. Teams will answer questions and/or complete tasks to assess any of the following topics:
 - a. Water Cycle, including
 - i. Processes of evaporation, condensation, precipitation, and run off
 - ii. Types of precipitation
 - iii. Location of major sources of water and the water cycle
 - b. Clouds, including
 - i. Types of clouds - identification from images or descriptions. Cloud types are limited to cirrus, cirrostratus, cirrocumulus, altostratus, stratus, stratocumulus, nimbostratus, cumulus, mammatus, lenticular, fog, and contrails.
 - ii. Processes of cloud formation
 - iii. Cloud relationships to weather conditions
 - c. Identification and description of the function of weather instruments (thermometer, barometer, rain gauge, hygrometer, sling psychrometer, wind vane, anemometer, weather balloon, radar, satellite)
 - d. Weather forecasting - using weather data to identify, describe, and/or predict weather conditions. To include:
 - i. Description/definitions of weather-related terms (dew point, front, humidity, relative humidity, pressure, air mass, windspeed, etc.)
 - ii. Identification of weather station and weather map symbols
 - iii. Reading weather maps/weather station data
 - iv. Using weather maps/weather station data to describe expected weather conditions
 - v. Using weather maps/weather station data over time to predict weather
 - e. Severe Weather in Florida - Hurricanes
 - i. Storm categories
 - ii. Parts of a storm
 - iii. Cone of uncertainty, prediction models, etc
 - iv. Weather phenomenon that lead to hurricanes
 - f. Preparation for Severe weather
 - i. Understand the need for a preparedness plan
 - ii. Be aware of common precautions for severe weather; ie lightning, tornado warning, hurricane preparedness, etc.

RELATED FLORIDA STANDARDS:

- .SC.5.E.7.1 Create a model to explain the parts of the water cycle. Water can be a gas, a liquid, or a solid and can go back and forth from one state to another.
- SC.5.E.7.2 Recognize that the ocean is an integral part of the water cycle and is connected to all of Earth's water reservoirs via evaporation and precipitation processes.
- SC.5.E.7.3 Recognize how air temperature, barometric pressure, humidity, windspeed and direction, and determine the weather in a particular place and time.
- SC.5.E.7.4 Distinguish among the various forms of precipitation (rain, snow, sleet, and hail), making connections to the weather in a particular place and time.
- SC.5.E.7.7 Design a family preparedness plan for natural disasters and identify the reasons for having such a plan.

SCORING:

- High score wins.
- Points will be awarded for the accuracy of responses.
- Ties will be broken by the accuracy or quality of preselected questions chosen by the event supervisor.